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Paleomagnetism of Lower Cretaceous rocks from east of the Liquine–Ofqui fault zone, southern Chile: evidence of small in-situ clockwise rotations

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ABSTRACT

One of the most conspicuous tectonic features of southern Chile is a 1000-km-long, N–S trending lineament known as the Liquiñe–Ofqui fault. Although generally regarded as a strike-slip fault, the amount and timing of strike-slip motion are unknown, and even the full extent and exact geometry of the feature remain in doubt. As part of ongoing investigations by Chilean and U.S. scientists we undertook a paleomagnetic study of Neocomian volcanosedimentary (Alto Palena Formation) and intrusive (North Patagonian batholith) rocks located just east of the main trace of the fault. All sites have normal polarity and form a common circular cluster. Indirect evidence strongly suggests extensive remagnetization of the Alto Palena rocks by intrusion of the batholith during the mid-Cretaceous long normal period. A paleomagnetic pole calculated from combined Alto Palena and North Patagonian batholith sites lies at 84.2°S, 159.9°E ($\alpha_{95} = 5.5°$; $k = 50$). when compared with Cretaceous reference poles for the South American craton this suggests a small but significant amount of clockwise rotation but little or no poleward transport. These results are consistent with other paleomagnetic data for the Chilean Andes and point to widespread rotational shear associated with the Liquiñe–Ofqui fault.

1. Introduction

A conspicuous difference between the Chilean Andes and the North American Cordillera is that the Andes appear to totally lack Mesozoic–Cenozoic displaced or exotic blocks [1,2] whereas the Cordillera has many such features. The two regions are similar in that paleomagnetic studies show that a pattern of small to moderate clockwise rotations is present in each. One possible explanation for the Chilean results is that the rotations are a response of small crustal blocks to dextral shear parallel to the continental margin, the ultimate cause of which is oblique subduction [1]. Other models for the mechanics of rotation have been proposed [3–5]. Whatever the explanation for the rotations, they probably are linked to the very long margin-parallel fault zones that are the dominant features of the Chilean coastal and main ranges [6]. Two major "trench-linked" strike-slip faults [7] have been recognized in the Chilean Andes, the Atacama fault zone [8–10] and the Liquine–Ofqui fault zone (LOFZ) [11]. The latter has been studied intensively during the last few years [6,12–18], although most of the effort has been concentrated on geochronology and reconnaissance geological mapping. In the single published paleomagnetic study of the area [6], a pattern of variable amounts of clockwise rotation found in the Lake District of southern Chile was interpreted as being related to the LOFZ.

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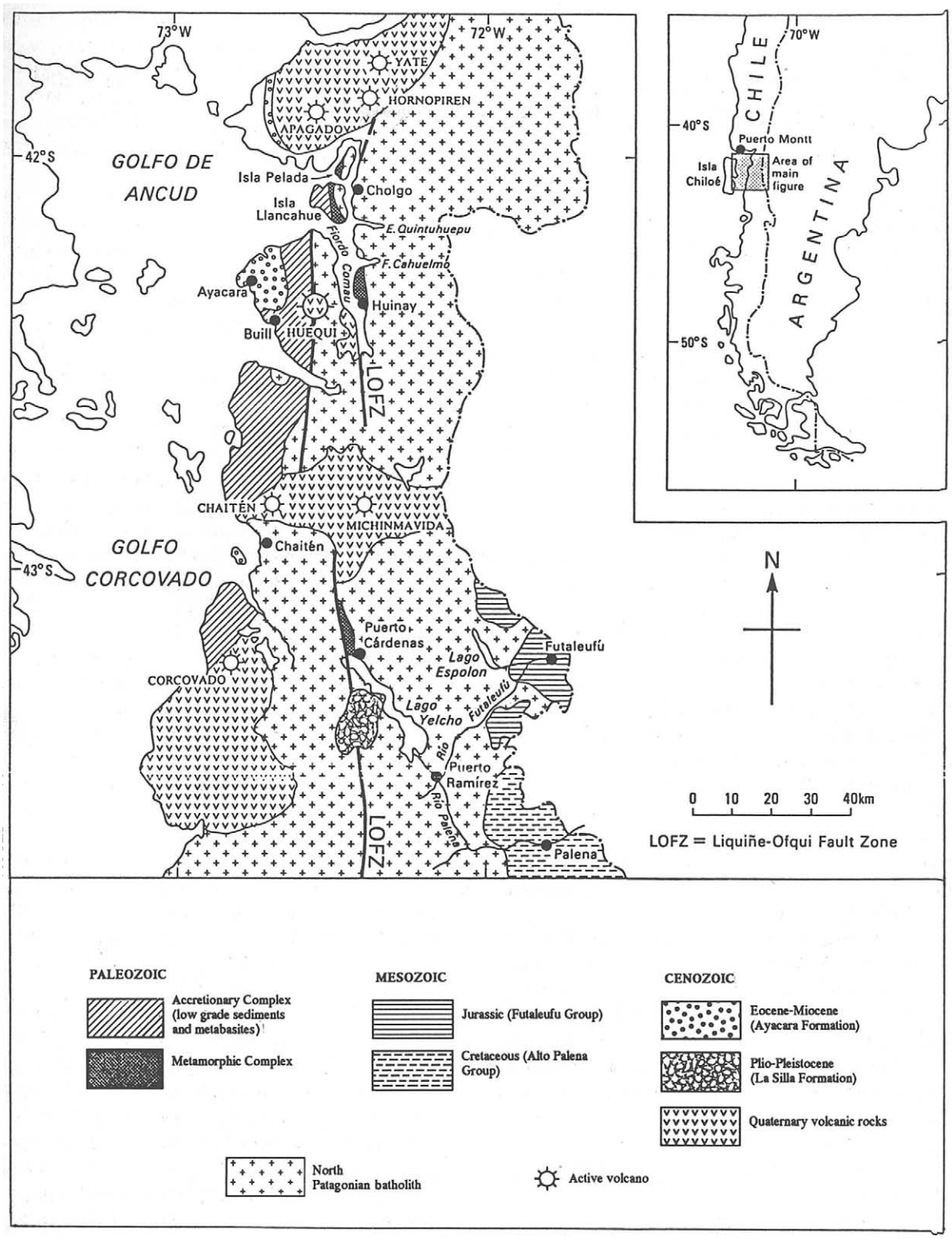


Fig. 1. Regional geological map of the province of the Palena, southern Chile. Modified after [16].

In the present study we have continued to search for rotations and displacements associated with the LOFZ. Initially we planned to sample rocks of early Cretaceous age thought to be present on both sides of the fault zone. However, as the study progressed it became clear that the age of the "early Cretaceous" rocks west of the LOFZ is very poorly constrained, and that the rocks themselves tend to be poor recorders of the ancient geomagnetic field. Therefore, in this paper we shall present only results from east of the fault.

2. Regional geological setting

In contrast to the central Andes, where episodic eastward migration of the magmatic arc and a well-developed forearc fold and thrust belt are present, the Andean Cordillera between 41°S and 49°S shows a unique Meso-Cenozoic tectonic evolution characterized by: (1) an essentially static magmatic arc, represented by the Patagonian batholith, (2) a relatively small amount of crustal shortening, and (3) an intra-arc strike-slip fault zone [2,19]. The Liquiñe–Ofqui fault zone is the most conspicuous tectonic element in the region. The zone probably developed along thermally weakened crust within the magmatic arc, as has been proposed for other major trench-parallel strike-slip faults [20–22]. We speculate that the LOFZ has accommodated shear arising from oblique subduction of the Nazca plate, and also has partly controlled the location of the Cenozoic magmatic arc.

In the region between 42°S and 44°S three distinct lithologic belts parallel the LOFZ (Fig. 1). These are, from west to east, the Coastal Range Paleozoic forearc accretionary complex [23,24], the North Patagonian batholith, and a sequence of Jurassic to Lower Cretaceous volcanic and volcanogenic sedimentary rocks.

The accretionary complex consists of low-grade metasedimentary rocks and interbedded tholeitic basalts. In the east these are intruded by the North Patagonian batholith (NPB), giving rise to isolated exposures of medium to high-grade metamorphic rocks. The NPB is composed of several distinct belts that grade eastward from mostly Mio-Pliocene granodiorites and tonalites to Jurassic–Lower Cretaceous monzogranites [16]. The western margin of the NPB is cut by the main trace of the LOFZ, along which younger intrusive bodies have been emplaced syntectonically. Isolated outcrops of Eocene to Miocene volcanosedimentary rocks (the Ayacara Fm. of Levi et al. [25]) are found along the coast of the Chiloé mainland, west of the LOFZ, where they are intruded by the NPB.

Rocks of the third (easternmost) belt include the Jurassic Futaleufu Group and Lower Cretaceous Palena Group [26]. These rocks, which are dominantly subaerial and marine volcanics, may represent a short-lived backarc basin [19]. The western flank of this volcanic belt is intruded by the NPB.

3. The Liquiñe–Ofqui fault zone

The LOFZ [27–29] is marked by a strikingly continuous set of aligned glacial valleys and fjords, along which isolated exposures of cataclastic and mylonitic rocks are found. The fault zone extends for about 1000 km along the main Cordillera, from 39°S near the small town of Liquiñe to the Golfo de Penas at about 47°S (Figs. 1 and 2). It has been divided into three segments, based on structure and other geological characteristics [11].

From 39°30′S to 41°30′S the LOFZ appears to be a single linear trace that trends N10°E. A belt of cataclastic and mylonitic rocks developed from older granitic rocks is present along the northernmost section of the zone. These are crosscut by an undeformed 28 Ma (K/Ar) dacite porphyry [19]. Right-lateral movements are suggested by conjugate mesoscopic faults within the cataclastic and mylonitic zone [30]. Between 41°30′ and 42°S the LOFZ juxtaposes Upper Miocene and Lower Cretaceous intrusive rocks [17] and shows less well developed cataclastic features [12].

From 42°S to 43°15′S the LOFZ runs through both the North Patagonian batholith and the accretionary complex, generating discontinuous strips of cataclastic and mylonitic rocks. Field relationships and petrographic studies suggest that Miocene to Pliocene tonalites were intruded along a ductile dextral shear zone [18]. After the Late Miocene, fault displacement apparently had a significant vertical component [15]. Quaternary volcanic centers are also aligned along the north-

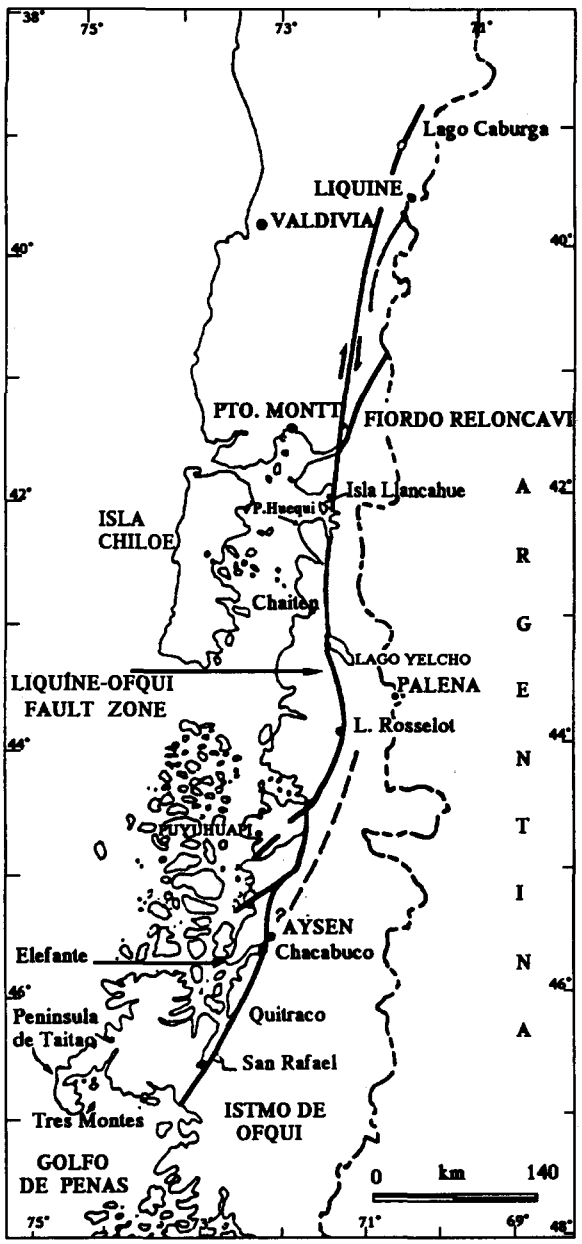


Fig. 2. Main trace of Liquiñe–Ofqui fault. After [11].

ern half of the LOFZ. These show no evidence of post-volcanic faulting.

The southern segment of the LOFZ is less well known. It appears as a set of *en-echelon* faults that defines a "scallop" pattern concave to the west. Generally, the structure is expressed by a regional pattern of linear geographic features, some with associated cataclastic rocks. The most conspicuous linear feature is a 2–3 km wide glacial valley that runs N–S from Lago Yelcho to Puyuhuapi (Fig. 2). This is partially filled by Paleogene(?) volcanic and volcanogenic sedimentary rocks (Las Juntas strata) and Neogene lacustrine sediments (La Silla strata) [31]. These rocks are capped by a 1.2 Ma basalt flow [15] which overlaps and seals the LOFZ trace. At Golfo de Penas, possibly a pull-apart basin related to the LOFZ [5], the sedimentary record appears to start in the Eocene.

The tectonic significance of the Liquiñe–Ofqui fault zone has been discussed by several authors. Some [1,30] speculate that it was caused by dextral shear related to oblique subduction of the Nazca (Farallon) plate beneath South America. Plate reconstructions [32–34] call for a period of highly oblique NE-directed subduction from around 50 Ma to 26 Ma, roughly coeval with the general timing for the strike-slip motion on the LOFZ. The subsequent gradual change to a predominant dip-slip (normal) component after the Miocene [15,30] probably represents a response to nearly orthogonal Nazca–South America convergence that followed breakup of the Farallon plate at 26 Ma.

An alternative model [5] views the LOFZ as a result of collision of a "rigid indenter" (the Chile Rise) with the Peru–Chile trench. Both indenter and oblique subduction models would produce dextral shear in the South American continental margin north of the Golfo de Penas.

An important enigma posed by the LOFZ is why it shows so little clear evidence of lateral offset. One reason could be that evidence of strike-slip displacement is hidden beneath late Cenozoic strata or obscured by Mio-Pliocene plutons. Another possibility is that strain along the shear zone has been partitioned among secondary splays off the main fault trace, and that much of it has been taken up by block rotations and crustal thickening. Given our current knowledge of the geometry of the LOFZ, which at most places is represented by a single trace, major displacements on the fault obviously encounter a serious space problem. Because it is sensitive to both N–S displacements and local vertical-axis rotations, paleomagnetism is a good tool with which to address some of these problems.

4. Geology of the sampling area

The sampling area (Fig. 2) contains exposures of the North Patagonian batholith (NPB) in the west and Lower Cretaceous Palena Group [35,36] volcanics and volcano sedimentary rocks in the east. At this latitude the NPB consists primarily of biotite monzonite and monzogranite. These yield Rb/Sr whole-rocks isochrons ranging from 120 to 100 Ma, in agreement with stratigraphical constraints on their emplacement [16].

The Palena Group is composed of the Neocomian Alto Palena F (APF) and the Albian–Aptian Cordon de Las Tobas F [26]. We sampled only the former.

The APF has been divided into three members on the basis of lithofacies and fossil content [26]. The lower member consists of 250 m of marine shale, sandstone, conglomerate and sedimentary breccia. The middle member is dominantly clastic sedimentary rock interbedded with tuff and volcanic breccia, giving way upward to exclusively pyroclastic rocks. It reaches 830 m in thickness. The upper member, which is 160 m thick, is composed of shale and calcareous sandstone with minor tuffaceous intercalations. Fossil content indicates ages of lower Valanginian to upper Hauterivian [26].

A simple structure consisting of wide, gentle folds superimposed on intense extensional block

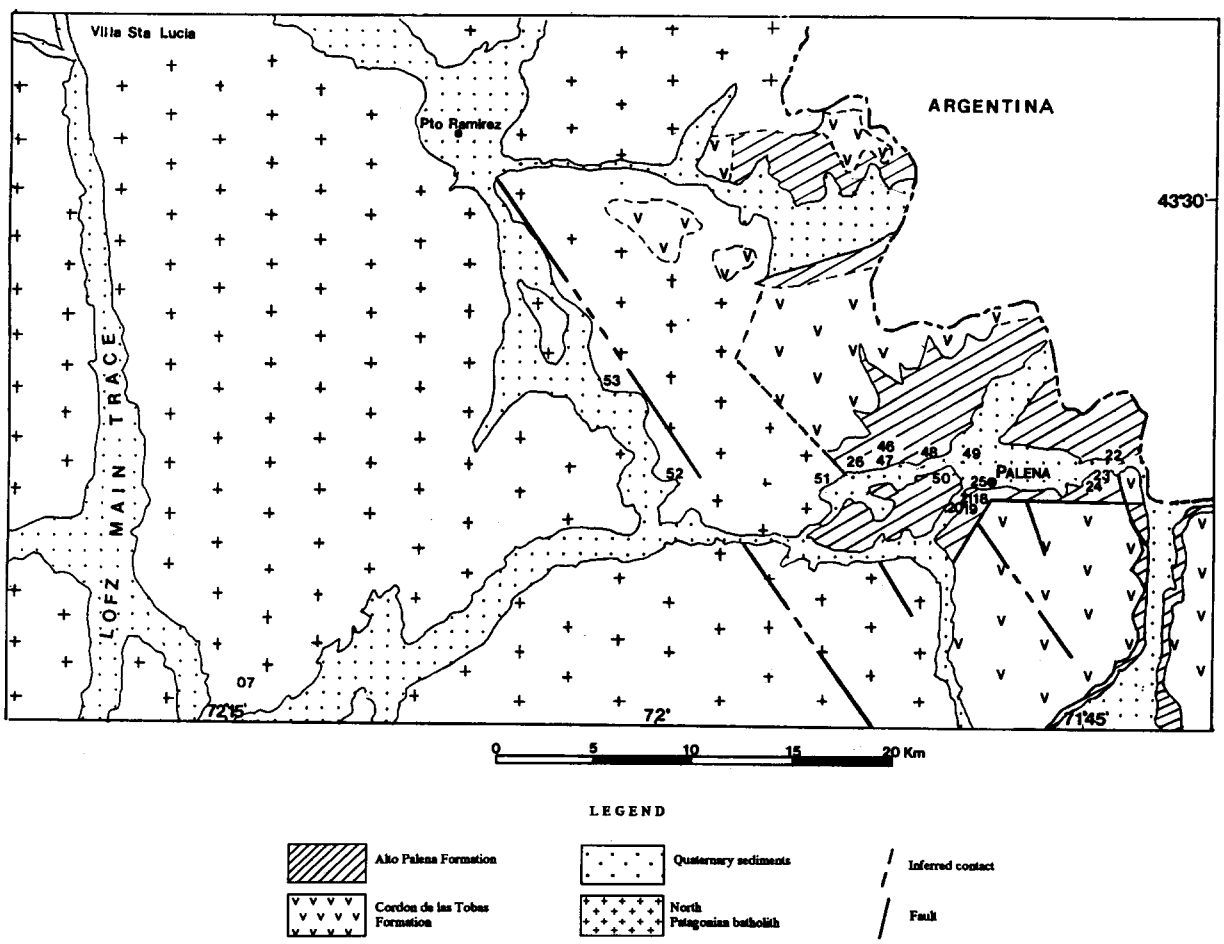


Fig. 3. Geologic map of Liquiñe–Ofqui fault zone. From [35,36]. Numbers indicate paleomagnetic sampling sites.

tectonics characterizes the region. Block tectonism is indicated by horst and graben structures defined by regional NW-striking faults and lineaments [26]. The absolute timing of these deformational events is unknown.

5. Paleomagnetism

In 1989 and 1990, 148 oriented samples were collected at nineteen sites near Palena (Fig. 3); fifteen sites are from the Alto Palena F and four are from the NPB. These were drilled and oriented *in situ* using standard techniques. Both sun and magnetic compass were used for orientation.

To determine blocking temperature and coercivity spectra of the natural remanent magnetization, two pilot specimens were obtained from a representative sample from each site. One specimen of each pair was thermally demagnetized to a maximum of 680°C and the other was demagnetized in an alternating magnetic field (AF), using a two-axis reciprocating tumbler, to a maximum of 100 mT.

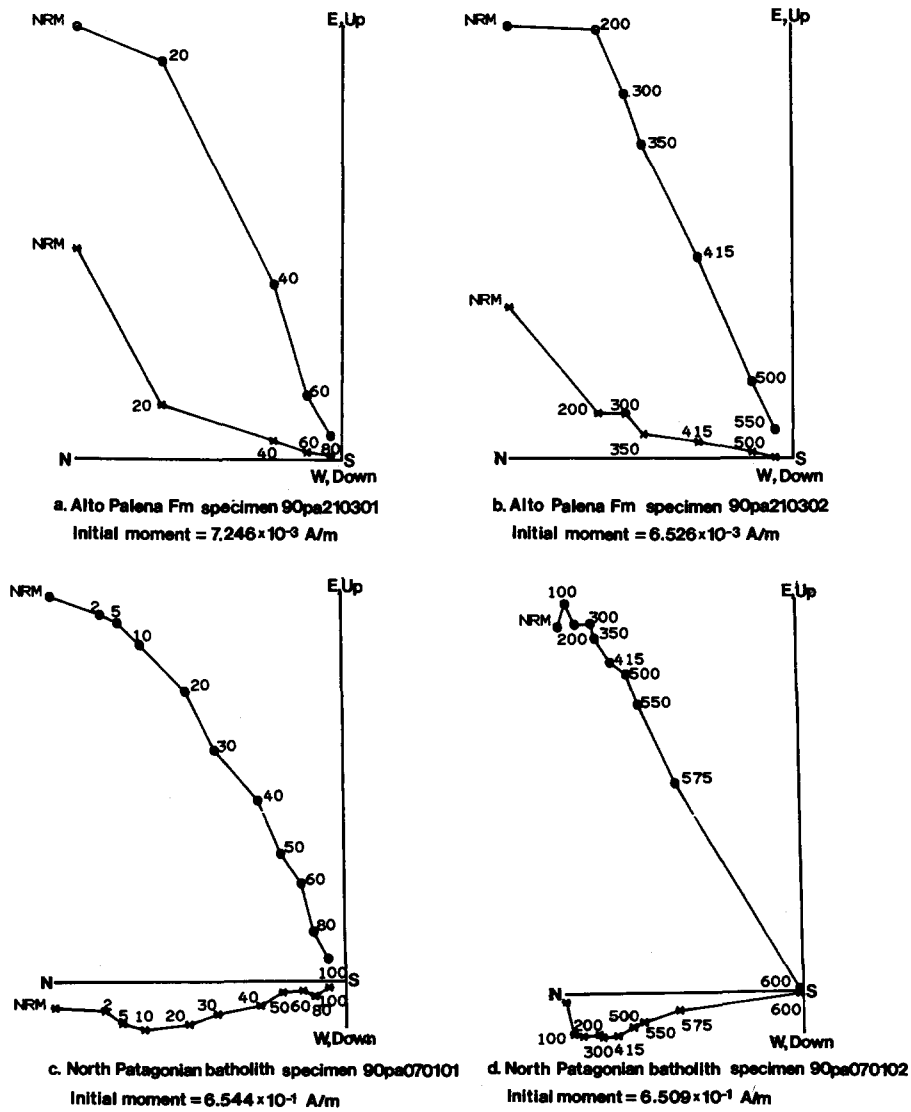


Fig. 4. Typical demagnetization diagrams. (a and b) Alto Palena Fm. (c and d) North Patagonian = Batholith. Not corrected for tilt. × = map view; ○ = vertical section.

Representative vector demagnetization diagrams [37] from the Alto Palena Fm. are shown in Figs 4a and b. AF demagnetization at peak fields above 20mT produced a demagnetization path that trended towards the origin, indicating isolation of a single component of magnetization; as Figs. 4a and b show, similar directions are removed by AF demagnetization from 20 mT to 80 mT and by thermal demagnetization from 350° to 550°C. This suggests that the stable magnetization is carried by magnetite and/or titanomagnetite. The sole exception is site pa22, which retained measurable magnetization up to 675°C, indicating a magnetization that probably resides in hematite.

Representative vector demagnetization diagrams from NPB samples are shown in Figs 4c and d. A steady trend towards the origin is found by AF and thermal demagnetization above 20 mT and 300°C. Magnetizations survive to 100 mT but not above 600°C. Titanomagnetite and/or magnetite probably carry the stable magnetization in these intrusive rocks.

For most APF and all NPB samples the directions isolated by these two techniques (AF and thermal) were indistinguishable. Therefore, AF was used for the remaining samples, as it avoids heat-induced mineralogical changes, and is faster and easier to perform.

Principal component analysis was applied to obtain magnetic directions from the demagnetization data. Directions of anchored and free lines [38] were calculated for each sample, from 4- to 10-step demagnetization paths. Subsequent analyses were carried out using sample directions calculated both ways in order to test whether some undetected higher stability component was present. Most sample directions grouped well within site. However, within some sites, a single sample direction fell well outside an otherwise tightly clustered group. These were excluded if they were poorly determined directions (indicated by a maximum angular deviation [38] exceeding 10°) or had atypical magnetic characteristics (principally abnormally high intensities, which probably indicate lightning strikes).

TABLE 1

Site mean directions from Alto Palena Formation and North Patagonian batholith

Site	Uncorrected		Corrrected		N	R	α_{95}	k	Location	
	Dec.	Inc.	Dec.	Inc.					Lat. (S)	Long. (E)
Alto Palena Formation										
pa18	18.1	−61.9	6.0 *	−44.5 *	4	3.992	2.0	375	43.62	288.20
pa19	15.9	−63.4	6.0 *	−44.5 *	4	3.978	2.0	136	43.62	288.20
pa20	4.5	−59.9	359.5	−40.3	7	6.928	6.6	83	43.62	288.20
pa21	14.8	−64.0	6.0 *	−44.5 *	7	6.993	2.0	857	43.62	288.20
pa22	331.0	−39.6	–	–	6	5.924	8.3	66	43.60	288.28
pa23	334.6	−71.1	11.9	−59.4	5	4.979	5.5	190	43.60	288.27
pa25	67.0	−75.6	22.3	−63.0	6	5.962	5.9	131	43.60	288.20
pa26	1.9	−49.7	9.4	−55.3	6	5.951	6.7	102	43.62	288.13
fu46	10.2	−57.3	14.2	−67.1	6	5.954	6.5	109	43.60	288.15
fu47	16.5	−49.3	20.9	−58.8	6	4.956	6.3	113	43.60	288.15
fu48	354.5	−57.1	352.3	−67.0	6	4.965	5.6	143	43.60	288.17
fu50	11.5	−62.6	351.9	−62.9	6	5.970	5.2	167	43.63	288.18
North Patagonian batholith										
pa07	351	−63.1	–	–	7	6.979	3.6	286	44.05	287.73
fu51	358.1	−58.9	–	–	5	4.862	14.5	29	43.63	288.07
fu52	15.6	−60.8	–	–	5	4.930	10.2	57	43.63	288.07
fu53	3.9	−53.0	–	–	5	4.973	6.3	148	43.60	287.95

Dec. and Inc. = declination and inclination (positive downward) N = number of samples R = resultant of N unit vectors α_{95} = radius of circle of 95% confidence k = precision parameter Lat. and Long. = latitude and longitude of site location

* Sites pa18, pa19 and pa21 combined and averaged together

A circularity test [39) shows that within-site distributions are circular, hence mean directions were calculated using standard Fisher statistics [40] from anchored and free line directions. Very little difference was found ($< 5°$) between mean directions calculated from free versus anchored lines. Observed differences are not systematic, and hence are probably attributable to random esperimental error. Therefore, because higher precision is obtained using anchored lines that method was preferred.

6. Results

Alto Palena sites: Table 1 and Fig. 5 summarize the paleomagnetic results for the Alto Palena F. Figure 5a shows that, with the exception of sites 22, 23 and 25, site-mean directions uncorrected for tilt lie in a cluster. Among them, sites 18, 19 and 21 have very similar directions. Because these sites were taken from sequential flows in the same stratigraphic section, they may not represent true time-independent readings of the an-

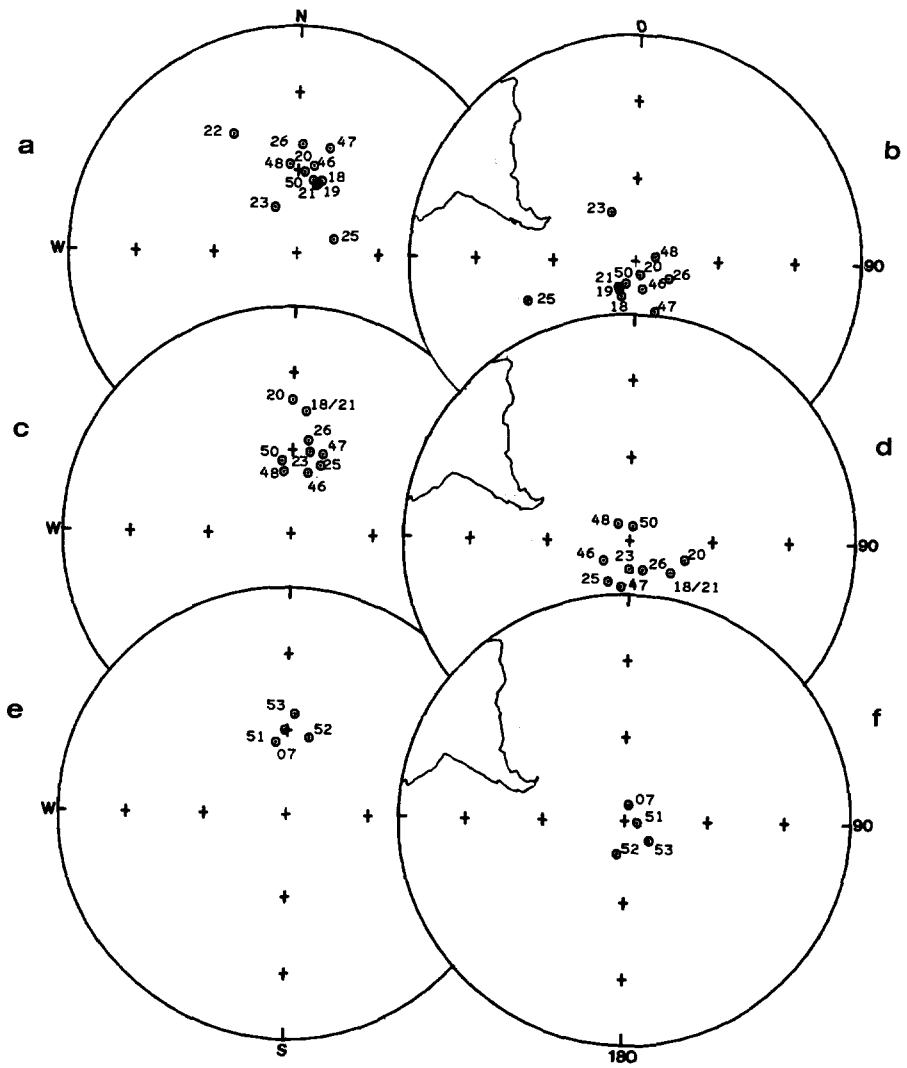


Fig. 5. Equal area projections of site-mean directions and corresponding virtual geomagnetic poles. (a and b) Alto Palena Fm. not corrected for tilt. (c and d) Alto Polera Fm. corrected for tilt. (e and f) North Patagonian batholith.

TABLE 2

Paleomagnetic poles calculated by assuming the end-member scenarios (see text). Reference poles are listed

Unit	N	R	k	α_{95}	Pole Lat. (S)	Pole Long. (E)
APF	9	8.806	41	8.1	83.1	166.7
NPB	4	3.963	81	10.3	86.1	134.7
NPB + APF *	13	12.540	26	8.2	85.3	193.1
NPB + APF	13	12.760	50	5.9	84.2	159.9
SG	54	52.400	33	3.4	83.8	94.6
PPB	18	17.460	32	6.3	78.7	358.4

Lat. and Long. = latitude and longitude of paleomagnetic pole, other abbrevations and symbols as in Table 1; APF = Alto Palena Formation, corrected for tilt; APF* = Alto Palera Formation uncorrected for tilt; NPB = North Patagonian batholith; SG = Serra Geral reference pole; PPB = Patagonian plateau basalt reference pole.

cient geomagnetic field, so samples from these three sites were averaged together to form a single "super-site". The only other modification to the data set was to discard site 22, which has an uncertain structural attitude, lies well outside the cluster of other site-mean directions, and is unique in its high apparent hematite content.

All sites have a circular distribution of sample directions, consistent with the hypothesis that only a single direction perturbed by random noise was sampled. Virtual geomagnetic poles (VGPs) were obtained from each site-mean direction, before and after correction for tilt, and averaged to yield poles for the APF (Table 2). As shown in Fig. 5c, correction for tilt produces a moderate improvement in grouping. This is more easily seen in plots of VGPs calculated from reliable site-mean directions, before and after correction (Fig. 5b and d). This suggests that the magnetization is pre-folding.

North Patagonian batholith sites: The results from our four sites in the easternmost NPB (Fig.

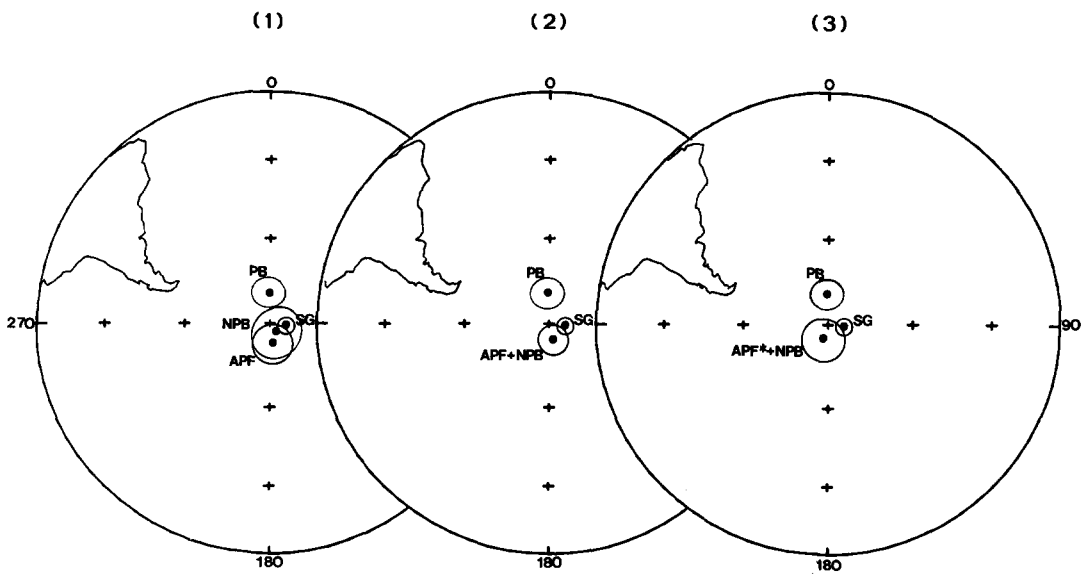


Fig. 6. Paleomagnetic poles calculated for three alternative scenarios (see text). (1) Primary magnetization: APF = Alto Palena Fm.; NPB = North Patagonian batholith. (2) Remagnetization prior to folding: APF + NPB = pole calculated by combining VGP for tilt-corrected APF and NPB. (3) Remagnetization after folding: APF* + NPB = pole calculated from VGP for APF not corrected for tilt and NPB. Reference poles: PB = Patagonian basalts [43], SG = Serra Geral [44].

5e and f, Table 1) should be regarded with caution for two reasons: (1) in general, a small number of sites does not properly average secular variation, and (2) intrusive rocks do not provide paleohorizontal control. However, the first consideration may not be very critical, because sites from plutonic rocks cool so slowly that they may average much of the secular variation within-site. A favorable consequence of this is that fewer sites are required to average secular variation adequately. Thus, the fairly circular, tight distribution of site-mean directions (Fig. 5e) may indicate adequate sampling of secular variation.

The second consideration is more serious. As with most batholiths, tilting of the North Patagonian batholith cannot be easily assessed. Therefore, directions from these intrusive rocks may be erroneus because of undetected post-magnetization tilt. However, close agreement between mean directions for widespread sites within the batholith, and between batholith mean directions and directions from the Alto Palena Fm., argues that tilting probably is minimal.

7. Discussion

Before we can consider the possible tectonic significance of our results the age and origin of the magnetizations must be discussed.

The Alto Palena Fm. and North Patagonian batholith sites all have normal polarity. Because a considerable thickness of APF was sampled, and because acquisition of TRM by the slow-cooling NPB presumably took a long time, it seems fairly certain that all our sites acquired their magnetization during a time when the geomagnetic field remained normal for a long time. The obvious candidate is the Cretaceous long normal interval, roughly from 125 to 83 Ma [41]. If this is the case then the APF, which has a well-documented Neocomian (roughly 145–125 Ma) age [26], must have been remagnetized after deposition, presumably by intrusion of the Albian-Aptian (118–97.5 Ma) NPB [16]. However, other scenarios are possible. We examine three: (1) APF rocks retain a primary magnetization acquired during one of the several short normal polarity intervals that occurred during its deposition in the Neocomian, (2) APF rocks were remagnetized by the NPB prior to folding, and (3) APF rocks were remagnetized by the NPB after folding.

To investigate these possibilities, paleomagnetic poles were calculated from site VGPs in a way that was appropriate for each (Table 2). For alternative (1), which assumes two different ages of magnetization for the APF and NPB, poles were calculated from each group separately. For alternatives (2) and (3), which assume a common age of magnetization, a single pole was calculated by combining APF and NPB data. As shown by Fig. 6, poles for the different alternatives are indistinguishable at the 95% confidence level. This shows that, regardless of the nature and precise timing of the magnetization, any of these poles can serve as a Lower Cretaceous pole for the study area. In other words, we are lucky; our tectonic conclusions are essentially independent of the magnetization history of the rocks. However, we would still like to determine which is correct.

Primary magnetization during the Neocomian appears to be the least likely. Assuming that the Neocomian geomagnetic polarity was reverse 50% of the time, the probability of randomly selecting ten samples that are exclusively normal is less than 0.1%. Of the two remagnetization scenarios, pre-folding remagnetization appears to be the more likely because correction for tilt reduces scatter. Also, dispersion of VGPs from the APF is smaller than is typical for full secular variation [42], consistent with some secular variation having been smoothed within-site. However, the circular distribution and total dispersion of VGPs from combined APF and NPB sites suggest that the full secular variation was adequately sampled (Fig. 5d and f).

From the discussion above it follows that tectonic interpretation should be made assuming extensive remagnetization of the APF by the NPB before folding. In using the APF + NPB pole, two suppositions are required: (1) APF rocks were flat-lying when remagnetized, and (2) APF and NPB magnetizations were acquired during or shortly after the Albian–Aptian NPB intrusive event, at about 100 Ma. Despite the loss of paleohorizontal control because of remagnetization, the agreement of APF and NPB directions over the entire sampling area implies that post-magnetization tilt is unlikely. Thus we regard the APF

+ NPB pole as reliable and suitable for comparison with available reference poles for the South American craton.

Unfortunately, there is no reliable mid-Cretaceous (100 Ma) reference pole for South America [43]. However, the recalculated [1] 128 Ma Serra Geral (SG) pole [44] and the 72 Ma Patagonian Plateau Basalts (PB) pole [43] (both in Table 2) define a short APW track along which the desired reference pole probably lies. To test this we rotated the mid-Cretaceous reference pole for North America [45] into the South America framework using the available reconstruction parameters [46]. This rotated pole lies on the 128 to 72 Ma track. Likewise, we rotated Lower Cretaceous and Upper Cretaceous African reference poles [47] into a fixed South America frame using the appropiate Euler poles [48]. These poles also lie along the SG–PB path. We conclude from this that relative motions of the sampling area can be determined by comparing the APF + NPB pole with SG and PB reference poles, using standard methods [49]. This calculation shows, that there is a small but significant (at 95% confidence) clockwise rotation of the area, $7.7 \pm 7.0°$ relative to the SG pole, or 21.4 ± 9.6 relative to the PB pole. Poleward transport of $2.2 \pm 5.8°$ and -6.6 ± 6.9, respectively, are not significant.

Some general interpretations of these results can be made within the context of the tectonic evolution of this part of the Andes:

(1) These Lower Cretaceous rocks have paleomagnetic inclinations that are essentially concordant, showing that they are neither an exotic terrane (unless they moved directly from the west) nor a far-traveled piece of the South American continental margin.

(2) The small clockwise rotation found is consistent with widespread evidence of such rotations elsewhere in the Andes south of the Arica deflexion [4,50–52]. While most Chilean paleomagnetic data have come from the Andean forearc, the results presented here are from rock units in a back-arc position, suggesting that the driving mechanism for these small rotations may operate throughout the orogen.

Oblique subduction of the Farallon (Nazca plate) beneath the South American plate during the Meso-Cenozoic appears as a common feature overall and can explain the occurrence of consistent rotations over much of the Chilean Andes. However, in contrast to the widespread evidence of latitudinal translation found in the North American Cordillera, detectable amounts of coastwise transport are absent. As argued elsewhere [1,51] this may be the result of differences in the outline of the western edges of the two continents.

Despite intensive geological study of the LOFZ, at the moment we cannot demonstrate the strike-slip motion or the causally related block rotations within the fault zone itself. However, the existence of sporadic outcrops of mylonitic and cataclastic rocks with N–S subhorizontal lineations parallel to the LOFZ is evidence that lateral motion has occurred. Obviously, the next step is to look for evidence of block rotations and tectonic transport west of the fault. What seems fairly clear from combining paleomagnetic [6, this work] and geological [15–16] evidence is that the LOFZ is a discrete zone of simple shear. We suspect that it represents part of the continent's response to NE-directed underthrusting by the Nazca plate, but this still remains to be demonstrated.

8. Summary and conclusions

(1) Twelve sites from the Alto Palena Fm. preserve a stable single-component magnetisation. Grouping and circularity of site-mean direction and VGPs improve upon correction for tilt, suggesting that the magnetization was acquired prior to folding. All sites from the NPB gave reliable stable magnetizations which are indistinguishable from the APF direction. The fact that polarities are entirely normal makes a Neocomian primary magnetization scenario for the APF unlikely; extensive remagnetization by the Albian–Aptian North Patagonian batholith during the Cretaceous long normal interval is more likely.

(2) Assuming that APF rocks remained flat through remagnetization and that the NPB has not undergone significant tilting, the paleomagnetic pole for the combined data sets lies at 84.2°S, 159.9°E ($\alpha_{95} = 5.5°$). Comparing this to reliable Cretaceous reference poles yields a small but significant clockwise rotation and negligible N–S translation at the 95% confidence level.

(3) The small clockwise rotations found in this study are similar to those found throughout much of the Chilean Andes. Oblique subduction is a possible common explanation for this phenomenon. Anomously large clockwise rotations (> 40°) that have been reported from rocks located in the vicinity of the main trace of the LOFZ at 41°S [6]. Variations in amounts of rotation near the LOFZ have been correlated with distance from the main trace of the fault [6]. We speculate that latitudinal changes in the overall geometry of the LOFZ are likely to control strain partitioning. This hypothesis will be addressed in a future paper.

(4) Geological studies by several authors [14,15,17,30] have led to the speculation that dextral shear has occurred within the LOFZ. This paleomagnetic study was unable to provide independent evidence of lateral motion along the segment of the LOFZ between 43°S and 44°S, although it does demonstrate that the eastern bounding block has not moved significantly with respect to the craton. Ongoing paleomagnetic sampling, accompanied by radiometric dating of outboard rocks, may provide more information about possible N–S translation and block rotation patterns.

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